

**STMIK KAPUTAMA BINJAI**  
**GOOGLE PROMPT ENGINEERING COURSE**



**PROJECT PROPOSAL**

**NAZSYA AI TUTOR**

**AI Smart Tutor Using Advanced Prompt Engineering Techniques**

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# PROJECT PROPOSAL

## 1. Project Title

### **Nazsya AI Tutor: AI Smart Tutor Using Advanced Prompt Engineering Techniques**

Nazsya AI Tutor is an AI-powered educational assistant designed to help learners understand topics through structured explanations, personalized study plans, interactive quizzes, and AI-based answer evaluation. The system applies advanced prompt engineering techniques to improve the clarity, structure, personalization, and educational usefulness of AI-generated responses.

## 2. Background

The rapid development of Artificial Intelligence has created new opportunities in the education sector. AI-powered learning systems can support students by providing explanations, examples, learning plans, quizzes, and feedback in a more flexible and interactive way.

However, general AI responses are often inconsistent, too broad, or not adapted to the learner's level. Without proper prompt design, AI-generated answers may lack structure, clarity, and educational focus. This can reduce the effectiveness of AI as a learning assistant.

Prompt engineering plays an important role in improving the quality of AI-generated responses. By designing clear and structured prompts, the AI model can be guided to produce more organized, relevant, and useful educational outputs.

Therefore, this project proposes the development of **Nazsya AI Tutor**, an AI-powered smart tutor that uses advanced prompt engineering techniques to generate structured learning responses. The system is designed to provide explanations based on learner level, generate personalized study plans, create interactive quizzes, and evaluate student answers using AI-based feedback.

## 3. Problem Statement

Based on the background, several problems are identified:

1. General AI responses are often not structured enough for educational use.
2. AI-generated explanations may not match the learner's level of understanding.
3. Learners need more than explanations; they also need study plans, quizzes, and feedback.
4. AI-generated learning content needs better consistency and reliability.
5. Student answers need to be evaluated using clear educational criteria.
6. Prompt engineering techniques need to be applied and compared to improve AI response quality.

## 4. Project Objectives

The objectives of this project are:

1. To develop an AI-powered tutor application using Streamlit and Large Language Model API integration.
2. To implement basic and optimized prompts for educational response generation.
3. To generate structured explanations using prompt engineering techniques.
4. To provide learner-level adaptation for Beginner, Intermediate, and Advanced users.
5. To generate personalized study plans based on selected study duration.
6. To create interactive multiple-choice quizzes for learner self-assessment.
7. To implement an AI Answer Evaluator using rubric-based and role-based prompting.
8. To compare basic and optimized prompts to evaluate the impact of prompt engineering.
9. To document the prompt templates used in the application through a Prompt Template Library.

## 5. Project Scope

The scope of this project focuses on the development of an AI-powered educational assistant named **Nazsya AI Tutor**.

The system includes the following scope:

1. Development of a web-based application using Streamlit.
2. Integration with OpenRouter API using the DeepSeek Chat model.
3. Implementation of basic prompt and optimized prompt response generation.
4. Implementation of structured educational explanations.
5. Implementation of learner-level selection: Beginner, Intermediate, and Advanced.
6. Implementation of study duration selection: 3 Days, 5 Days, and 7 Days.
7. Implementation of evaluator role selection: Friendly Tutor, Strict Lecturer, and Professional Examiner.
8. Generation of interactive quiz questions using a structured quiz format.
9. Evaluation of student-submitted answers using AI-generated feedback.
10. Display of prompt engineering analysis and performance evaluation table.

This project does not include user account management, database storage, online deployment to a public server, official academic grading, or permanent storage of student essay submissions.

## 6. Tools and Technologies

The development of Nazsya AI Tutor uses several tools and technologies to support application development, AI integration, prompt engineering implementation, and user interface design.

| Tool / Technology | Function |
|-------------------|----------|
|-------------------|----------|

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| Python                | Main programming language for application development          |
| Streamlit             | Framework used to build the web-based user interface           |
| OpenRouter API        | API provider used to connect the application with the AI model |
| DeepSeek Chat         | Large Language Model used to generate AI responses             |
| OpenAI-Compatible SDK | Python library used to access OpenRouter API                   |
| python-dotenv         | Library used to load API keys from environment variables       |
| Visual Studio Code    | Code editor used for development                               |
| Figma                 | Tool used for UI design planning and interface prototype       |
| Microsoft Word / PDF  | Used for documentation and final report preparation            |

The final application is built using **Streamlit** and integrated with **OpenRouter API** using the **DeepSeek Chat** model. The API key is stored in a .env file to improve security and prevent direct exposure of sensitive credentials. This matches the final implementation in the application code, where the system loads OPENROUTER\_API\_KEY and connects to OpenRouter using the OpenAI-compatible client.

## 7. Prompt Engineering Techniques

This project applies several prompt engineering techniques to improve the quality of AI-generated educational responses. The techniques are used to make the AI output more structured, personalized, interactive, and suitable for learning purposes.

The prompt engineering techniques used in Nazsya AI Tutor include:

### 1. Basic Prompting

Basic prompting is used to generate a simple and general response from the AI. This prompt serves as a baseline for comparison with the optimized prompt.

### 2. Structured Prompting

Structured prompting is used to guide the AI to generate responses in organized sections, such as Definition, Detailed Explanation, Real-World Example, Summary, Study Plan, and Interactive Quiz.

### 3. Few-Shot Prompting

Few-shot prompting is applied by providing an example topic and answer format. In the application, the example topic “What is gravity?” is used as a formatting guide so the AI can follow a consistent educational structure.

### 4. Learner-Level Adaptation

The optimized prompt adapts the explanation based on the selected learner level: Beginner, Intermediate, or Advanced.

### 5. Study Plan Prompting

The AI is instructed to generate a personalized study plan based on the selected duration: 3 Days, 5 Days, or 7 Days.

### 6. Structured Quiz Generation

The quiz is generated using a fixed format with QUIZ\_START and QUIZ\_END

delimiters. This allows the system to extract quiz content and convert it into interactive multiple-choice questions.

#### **7. Role-Based Prompting**

The evaluator feedback style is adjusted based on the selected role: Friendly Tutor, Strict Lecturer, or Professional Examiner.

#### **8. Rubric-Based Evaluation**

The AI Answer Evaluator assesses student answers using three criteria: Accuracy, Completeness, and Clarity. Each criterion is scored from 0 to 5, and the total score is calculated out of 15.

#### **9. Prompt Optimization**

Prompt optimization is used to improve the quality of AI-generated responses by comparing the basic prompt and optimized prompt. This technique helps demonstrate that a more detailed and structured prompt can produce clearer, more complete, and more educationally useful responses.

### **8. System Features**

Nazsya AI Tutor includes several main features designed to support interactive and personalized learning.

#### **8.1 Smart Explanation**

The system generates educational explanations based on the topic entered by the user. The optimized response includes structured sections such as definition, detailed explanation, real-world example, summary, study plan, and quiz.

#### **8.2 Basic and Optimized Prompt Response**

The application generates two types of AI responses:

##### **1. Basic Prompt Response**

A short and general explanation generated using a simple prompt.

##### **2. Optimized Prompt Response**

A structured and personalized educational response generated using advanced prompt engineering techniques.

This comparison helps demonstrate the impact of prompt engineering on AI response quality.

#### **8.3 Learner Level Selection**

Users can select one of three explanation levels:

1. Beginner
2. Intermediate
3. Advanced

The selected level affects the depth, terminology, and complexity of the AI-generated explanation.

#### **8.4 Study Duration Selection**

Users can select the study duration:

1. 3 Days
2. 5 Days
3. 7 Days

The selected duration is used by the AI to generate a personalized study plan.

#### **8.5 Evaluator Role Selection**

Users can select the evaluator role:

1. Friendly Tutor
2. Strict Lecturer
3. Professional Examiner

The selected role affects the tone and style of AI feedback.

#### **8.6 Interactive Quiz**

The system generates three multiple-choice questions related to the learning topic. Users can answer the quiz directly in the application, and the system calculates the quiz score automatically.

#### **8.7 AI Answer Evaluator**

Users can write their own answer and submit it for AI evaluation. The AI evaluates the answer using Accuracy, Completeness, and Clarity criteria, then provides score, strengths, weaknesses, and suggestions for improvement.

#### **8.8 Prompt Engineering Analysis**

The application displays a prompt engineering analysis section that summarizes the techniques used in the system, including Structured Prompting, Role-Based Prompting, Few-Shot Prompting, Prompt Optimization, and Rubric-Based Evaluation.

### **9. Methodology / Development Plan**

The project development is carried out through several stages:

## **9.1 Project Planning**

The first stage is identifying the project idea, objectives, target users, and main features. The project focuses on building an AI-powered educational assistant that applies prompt engineering techniques.

## **9.2 Initial Prompt Exploration**

Several prompt variations are explored to understand how prompt design affects AI-generated responses. This includes basic prompts, structured prompts, learner-level prompts, and optimized prompts.

## **9.3 Application Development**

The application is developed using Streamlit. The interface includes a sidebar, feature cards, input section, learning session display, quiz section, and AI answer evaluator.

## **9.4 API Integration**

The application is integrated with OpenRouter API using the DeepSeek Chat model. The API key is stored in an environment variable using a .env file.

## **9.5 Prompt Engineering Implementation**

The system implements basic prompts, optimized prompts, quiz generation prompts, and evaluator prompts. The optimized prompt combines several techniques to generate structured and personalized learning responses.

## **9.6 Quiz Parsing and Interaction**

The AI-generated quiz is extracted using QUIZ\_START and QUIZ\_END delimiters. Each quiz line is split using the vertical line symbol |, allowing the system to display the questions as interactive radio buttons.

## **9.7 Answer Evaluation**

The AI Answer Evaluator allows users to submit their own answers. The system sends the answer to the AI model with an evaluator prompt and displays rubric-based feedback.

## **9.8 Testing and Refinement**

The application is tested using sample learning topics such as photosynthesis. The testing process evaluates response clarity, structure, quiz functionality, answer evaluation, and prompt comparison results.

## 9.9 Documentation

The project documentation includes the Project Proposal, Prompt Template Library, Final Project Report, Presentation Slides, and optional demo materials.

## 10. Expected Output

The expected outputs of this project are:

1. A working Streamlit-based AI tutor application named Nazsya AI Tutor.
2. AI-generated structured explanations based on user questions.
3. Personalized study plans based on selected study duration.
4. Interactive multiple-choice quizzes generated by AI.
5. AI-based answer evaluation using rubric-based and role-based prompting.
6. Basic and optimized prompt comparison results.
7. Prompt Engineering Analysis section in the application.
8. Prompt Template Library documenting the prompts used in the project.
9. Final Project Report documenting the complete development process.
10. Presentation Slides for final project explanation.
11. Optional demo video showing the application workflow.

## 11. Project Timeline

| Development Stage | Activity                                                                     |
|-------------------|------------------------------------------------------------------------------|
| Stage 1           | Project idea selection and planning                                          |
| Stage 2           | Initial prompt exploration and basic prompt testing                          |
| Stage 3           | Structured answer generator development                                      |
| Stage 4           | Multi-level explanation and study plan generator implementation              |
| Stage 5           | Interactive quiz generator development                                       |
| Stage 6           | AI Answer Evaluator implementation                                           |
| Stage 7           | Prompt comparison and optimization                                           |
| Stage 8           | UI design planning and interface refinement                                  |
| Stage 9           | Final testing and debugging                                                  |
| Stage 10          | Documentation, Prompt Template Library, report, slides, and demo preparation |

## 12. Conclusion

Nazsya AI Tutor is proposed as an AI-powered educational assistant that uses advanced prompt engineering techniques to improve the quality of AI-generated learning responses. The system is designed to provide structured explanations, personalized study plans, interactive quizzes, and AI-based answer evaluation.



By implementing basic prompting, structured prompting, few-shot prompting, learner-level adaptation, quiz generation, role-based prompting, and rubric-based evaluation, this project demonstrates how prompt engineering can improve the clarity, structure, personalization, consistency, and educational usefulness of AI responses.

The final output of this project is expected to be a functional AI tutor application supported by proper documentation, prompt templates, testing evidence, and presentation materials.